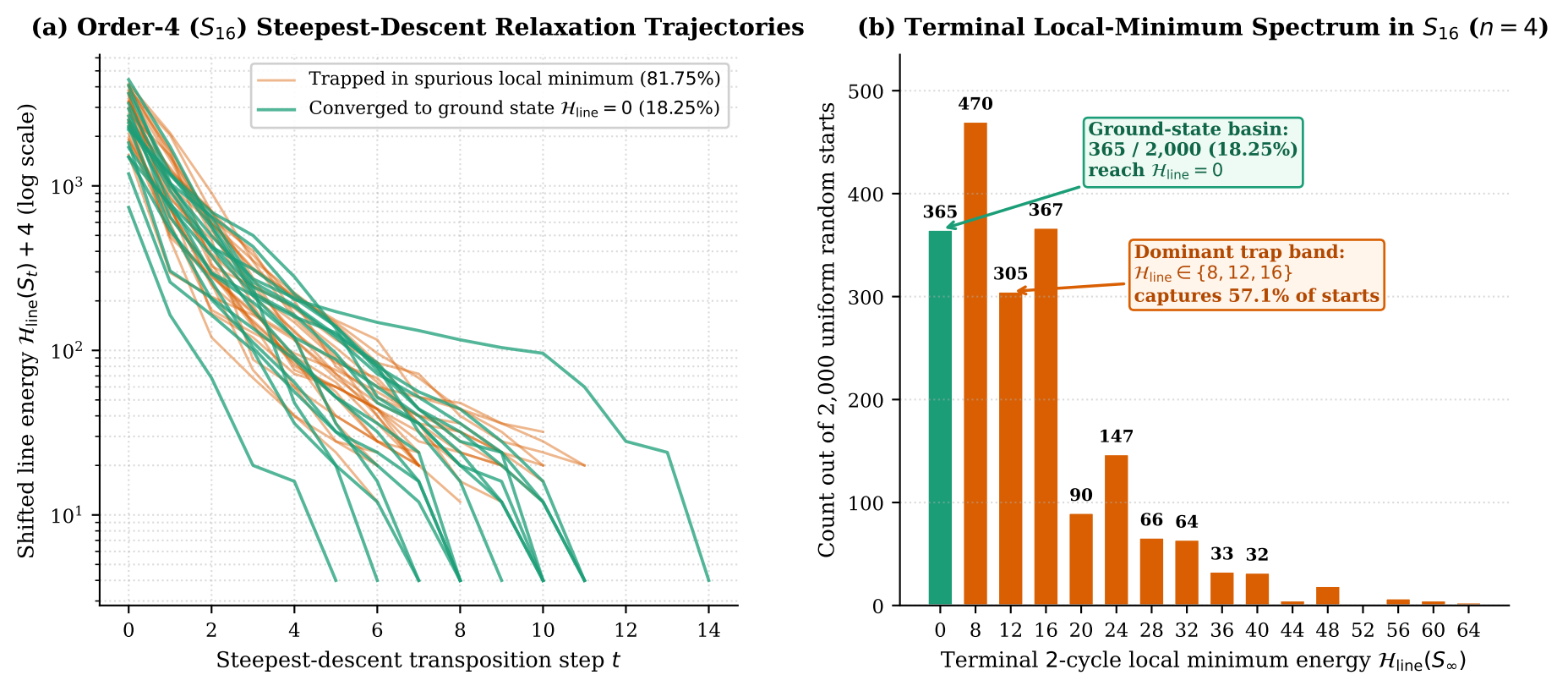
(a) Order-4 ($S_{16}$) Steepest-Descent Relaxation Trajectories
Trapped in spurious local minimum (81.75%)
Converged to ground state $\mathcal{H}_{\text{line}} = 0$ (18.25%)
Shifted line energy $\mathcal{H}_{\text{line}}(S_t) + 4$ (log scale)
$10^3$
$10^2$
$10^1$
0
2
4
6
8
10
12
14
Steepest-descent transposition step $t$
(b) Terminal Local-Minimum Spectrum in $S_{16}$ ($n = 4$)
Ground-state basin:
365 / 2,000 (18.25%)
reach $\mathcal{H}_{\text{line}} = 0$
Dominant trap band:
$\mathcal{H}_{\text{line}} \in \{8, 12, 16\}$
captures 57.1% of starts
Count out of 2,000 uniform random starts
0
100
200
300
400
500
470
365
367
305
147
90
66
64
33
32
0
8
12
16
20
24
28
32
36
40
44
48
52
56
60
64
Terminal 2-cycle local minimum energy $\mathcal{H}_{\text{line}}(S_\infty)$